

Phishing Email Detection Model

Project Overview

Build a machine learning model using Scikit-learn to detect phishing emails and classify them as safe or phishing.

Key Features

- Train on a dataset of phishing and legitimate emails
- Extract and analyze email features such as URLs and keywords
- Classify emails as Phishing or Safe
- Display accuracy and confusion matrix

Technologies Used

Python, Scikit-learn, Pandas, NumPy, Matplotlib

Folder Structure

```
phishing-email-detection/  
├──  
│   ├── phishing_detector.py  
│   ├── emails.csv  
│   ├── requirements.txt  
│   └── README.md
```

Install Required Packages

```
pip install pandas scikit-learn matplotlib numpy
```

phishing_detector.py

```
import pandas as pd  
from sklearn.model_selection import train_test_split  
from sklearn.feature_extraction.text import CountVectorizer  
from sklearn.naive_bayes import MultinomialNB  
from sklearn.metrics import accuracy_score, confusion_matrix  
import matplotlib.pyplot as plt  
  
# Sample dataset  
data = {  
    'email': [  
        'Win a free iPhone now',  
        'Your bank account is suspended',  
        'Meeting scheduled for tomorrow',  
        'Project report attached',  
        'Click here to claim reward'  
    ],  
    'label': ['phishing', 'phishing', 'safe', 'safe', 'phishing']  
}
```

```

}

df = pd.DataFrame(data)

# Features and labels
X = df['email']
y = df['label']

# Convert text into vectors
vectorizer = CountVectorizer()
X_vectorized = vectorizer.fit_transform(X)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X_vectorized, y, test_size=0.2, random_state=42
)

# Train model
model = MultinomialNB()
model.fit(X_train, y_train)

# Predictions
predictions = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, predictions)
print(f"Accuracy: {accuracy * 100:.2f}%")

# Confusion Matrix
cm = confusion_matrix(y_test, predictions)
print("Confusion Matrix:")
print(cm)

# Test custom email
test_email = ["Verify your account immediately"]
test_vector = vectorizer.transform(test_email)

result = model.predict(test_vector)

print(f"\nEmail Classification: {result[0]}")

```

Model Features

Feature	Description
Dataset Training	Trains model using phishing and legitimate emails
Feature Extraction	Analyzes email text and suspicious keywords
Email Classification	Predicts whether email is safe or phishing
Performance Metrics	Displays accuracy and confusion matrix

Expected Outcome

The model successfully classifies emails as Phishing or Safe with high accuracy based on textual content and URL features.

Conclusion

The Phishing Email Detection Model project demonstrates how machine learning can improve cybersecurity by detecting phishing emails. The system analyzes email content and predicts whether an email is safe or malicious using Scikit-learn algorithms.